

Probability & Statistics for Computer Scientists

Detailed Solutions: Chapters 3 & 4

Exercises 3.1 - 3.24, 3.28 | Exercises 4.1 - 4.5, 4.16 - 4.26

CHAPTER 3 — Discrete Random Variables and Their Distributions

Exercise 3.1

A computer virus corrupts two files independently. $P(\text{first}) = 0.4$, $P(\text{second}) = 0.3$. X = number of corrupted files.

(a) Probability Mass Function (PMF)

Using independence:

$$P(X=0) = (1-0.4)(1-0.3) = 0.6 \times 0.7 = 0.42$$

$$P(X=1) = 0.4 \times 0.7 + 0.6 \times 0.3 = 0.28 + 0.18 = 0.46$$

$$P(X=2) = 0.4 \times 0.3 = 0.12$$

x	P(X=x)	F(x) [CDF]
0	0.42	0.42
1	0.46	0.88
2	0.12	1.00

(b) CDF F(x)

$F(x)$ is constant between integer values; it jumps at each x by $P(x)$:

$$F(x) = 0 \quad \text{for } x < 0$$

$$F(x) = 0.42 \quad \text{for } 0 \leq x < 1$$

$$F(x) = 0.88 \quad \text{for } 1 \leq x < 2$$

$$F(x) = 1.00 \quad \text{for } x \geq 2$$

Exercise 3.2

Network blackouts per day. Each blackout causes \$500 loss. $L = 500X$.

x	P(x)	x*P(x)	x^2 * P(x)
0	0.7	0	0
1	0.2	0.2	0.2
2	0.1	0.2	0.4
Sum	1.0	0.4	0.6

$$E(X) = 0.4 \quad \text{Var}(X) = E(X^2) - [E(X)]^2 = 0.6 - 0.16 = 0.44$$

$$E(L) = 500 \times E(X) = 500 \times 0.4 = \$200$$

$$\text{Var}(L) = 500^2 \times \text{Var}(X) = 250,000 \times 0.44 = \$110,000$$

Expected daily loss = \$200; Variance = \$110,000

Exercise 3.3

5 blocks, 1 error in one block. Test 3 randomly selected blocks. X = errors found. This is hypergeometric: N=5 total, K=1 error block, n=3 selected.

$$P(X=0) = C(1,0) \cdot C(4,3) / C(5,3) = 1 \cdot 4 / 10 = 4 / 10 = 0.4$$

$$P(X=1) = C(1,1) \cdot C(4,2) / C(5,3) = 1 \cdot 6 / 10 = 6 / 10 = 0.6$$

$$E(X) = 0 \cdot 0.4 + 1 \cdot 0.6 = 0.6$$

$$E(X^2) = 0^2 \cdot 0.4 + 1^2 \cdot 0.6 = 0.6$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 0.6 - 0.36 = 0.24$$

E(X) = 0.6, Var(X) = 0.24

Exercise 3.4

Fair die: X takes values 1,2,3,4,5,6 each with probability 1/6.

$$E(X) = (1+2+3+4+5+6) / 6 = 21 / 6 = 3.5$$

$$E(X^2) = (1+4+9+16+25+36) / 6 = 91 / 6 = 15.1667$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 91 / 6 - (3.5)^2 = 91 / 6 - 12.25 = 35 / 12 = 2.9167$$

E(X) = 3.5, Var(X) = 35/12 = 2.917

Exercise 3.5

12 programs, 5 must be upgraded. 4 chosen for testing. Hypergeometric: N=12, K=5 upgradable, n=4. C(12,4) = 495.

(a) P(at least 2 must be upgraded)

$$P(X=0) = C(5,0) \cdot C(7,4) / C(12,4) = 1 \cdot 35 / 495 = 35 / 495$$

$$P(X=1) = C(5,1) \cdot C(7,3) / C(12,4) = 5 \cdot 35 / 495 = 175 / 495$$

$$P(X \geq 2) = 1 - 35 / 495 - 175 / 495 = 285 / 495 = 19 / 33 = 0.5758$$

P(X ≥ 2) = 19/33 ≈ 0.5758

(b) Expected number of upgradable programs in sample

$$E(X) = n * K/N = 4 * 5/12 = 20/12 = 5/3 = 1.667$$

$$E(X) = 5/3 \approx 1.667$$

Exercise 3.6

6 blocks, 1 error. Test 2 blocks randomly. X = errors found.

$$P(X=0) = C(1,0) * C(5,2) / C(6,2) = 10/15 = 2/3$$

$$P(X=1) = C(1,1) * C(5,1) / C(6,2) = 5/15 = 1/3$$

$$E(X) = 0 * (2/3) + 1 * (1/3) = 1/3$$

$$E(X) = 1/3 \approx 0.333$$

Exercise 3.7

Home runs per game: $P(0)=0.4$, $P(1)=0.4$, $P(2)=0.2$. Two independent games, Y = total home runs.

$$E(X) = 0*0.4 + 1*0.4 + 2*0.2 = 0.8$$

$$E(X^2) = 0*0.4 + 1*0.4 + 4*0.2 = 1.2$$

$$\text{Var}(X) = 1.2 - 0.64 = 0.56$$

$$E(Y) = E(X_1) + E(X_2) = 0.8 + 0.8 = 1.6$$

$$\text{Var}(Y) = \text{Var}(X_1) + \text{Var}(X_2) = 0.56 + 0.56 = 1.12 \quad (\text{by independence})$$

$$E(Y) = 1.6, \text{Var}(Y) = 1.12$$

Exercise 3.8

4 passwords, tries until correct (without replacement). X = number of wrong passwords tried. By symmetry, each of the 4 positions is equally likely to be the correct one. X can be 0, 1, 2, or 3, each with probability $1/4$.

x (wrong tries)	$P(X=x)$	$x * P(x)$	$x^2 * P(x)$
0	$1/4 = 0.25$	0	0
1	$1/4 = 0.25$	0.25	0.25
2	$1/4 = 0.25$	0.50	1.00
3	$1/4 = 0.25$	0.75	2.25
Sum	1.00	1.50	3.50

$$E(X) = 1.50$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 3.50 - 2.25 = 1.25$$

$$E(X) = 1.5, \text{Var}(X) = 1.25$$

Exercise 3.9

Download time X : $E(X) = 40$ sec, $\text{Std}(X) = 5$ sec. $P(X > 60)$?

Applying Chebyshev's inequality with $\epsilon = 60 - 40 = 20$:

$$P(X > 60) \leq P(|X - 40| > 20) \leq (\sigma/\epsilon)^2 = (5/20)^2 = 1/16 = 0.0625$$

$P(X > 60) \leq 6.25\%$ (using Chebyshev's inequality)

Exercise 3.10

Traffic accidents per day: $P(0)=0.6$, $P(1)=0.2$, $P(2)=0.2$. $P(\text{Friday} > \text{Thursday})$?

$$\begin{aligned} P(F > T) &= P(F=1, T=0) + P(F=2, T=0) + P(F=2, T=1) \\ &= 0.2 \cdot 0.6 + 0.2 \cdot 0.6 + 0.2 \cdot 0.2 \\ &= 0.12 + 0.12 + 0.04 = 0.28 \end{aligned}$$

$P(\text{Friday} > \text{Thursday}) = 0.28$

Exercise 3.11

Two dice: $X = \min$, $Y = \max$. If equal: $X = Y$.

(a) Joint PMF — $P(X=x, Y=y)$

For $x < y$: 2 outcomes give $(X=x, Y=y)$ out of 36, so $P = 2/36 = 1/18$.

For $x = y$: only 1 outcome (both dice show x), so $P = 1/36$.

X\Y	1	2	3	4	5	6
1	1/36	2/36	2/36	2/36	2/36	2/36
2	0	1/36	2/36	2/36	2/36	2/36
3	0	0	1/36	2/36	2/36	2/36
4	0	0	0	1/36	2/36	2/36
5	0	0	0	0	1/36	2/36
6	0	0	0	0	0	1/36

(b) Independence

X and Y are NOT independent. For example: $P(X=6, Y=1) = 0$, but $P(X=6) \cdot P(Y=1) > 0$. Also they are linked by $X \leq Y$ by definition.

(c) Marginal PMF of X

$$P(X=x) = P(\text{both} \geq x) - P(\text{both} \geq x+1) = [(7-x)/6]^2 - [(6-x)/6]^2:$$

x	P(X=x)
1	11/36
2	9/36
3	7/36
4	5/36
5	3/36

(d) $P(Y=5 | X=2)$

$$P(Y=5 | X=2) = P(X=2, Y=5) / P(X=2) = (2/36) / (9/36) = 2/9 = 0.222$$

$$P(Y=5 | X=2) = 2/9 \approx 0.222$$

Exercise 3.12

Joint distribution $P(x,y)$ with marginals $PX(0)=0.7$, $PX(1)=0.3$; $PY(0)=0.7$, $PY(1)=0.3$.

$x \setminus y$	$y=0$	$y=1$	$PX(x)$
$x=0$	0.5	0.2	0.7
$x=1$	0.2	0.1	0.3
$PY(y)$	0.7	0.3	1.00

(a) Independence of X and Y

$$PX(0) * PY(0) = 0.7 * 0.7 = 0.49 \neq P(0,0) = 0.5$$

X and Y are NOT independent.

(b) Independence of $(X+Y)$ and $(X-Y)$

Possible joint values of $(S=X+Y, D=X-Y)$: $(0,0)=0.5$, $(1,1)=0.2$, $(1,-1)=0.2$, $(2,0)=0.1$

$$P(S=0) * P(D=0) = 0.5 * (0.5+0.1) = 0.5 * 0.6 = 0.30 \neq P(S=0, D=0) = 0.50$$

$(X+Y)$ and $(X-Y)$ are NOT independent.

Exercise 3.13

Joint distribution: $P(0,0)=0.2$, $P(0,2)=0.3$, $P(1,1)=0.1$, $P(2,0)=0.3$, $P(2,2)=0.1$.

(a) PMF of $Z = X+Y$

$z = X+Y$	Pairs (X,Y)	$P(Z=z)$
0	(0,0)	0.2
2	(0,2), (1,1), (2,0)	$0.3+0.1+0.3 = 0.7$
4	(2,2)	0.1

(b) PMF of $U = X-Y$

$u = X-Y$	Pairs (X,Y)	$P(U=u)$
-2	(0,2)	0.3
0	(0,0), (1,1), (2,2)	$0.2+0.1+0.1 = 0.4$
2	(2,0)	0.3

(c) PMF of $V = XY$

$v = XY$	Pairs (X,Y)	$P(V=v)$
0	(0,0), (0,2), (2,0)	$0.2+0.3+0.3 = 0.8$
1	(1,1)	0.1
4	(2,2)	0.1

Exercise 3.14

Charge $Z = X*Y$ (hours x cents/hour). Joint distribution given.

Computing $Z = X*Y$ for each non-zero cell:

$Z = X*Y$	Contributing (X,Y) pairs	$P(Z)$
2	(2,1)=0.06; (1,2)=0.10	0.16
3	(3,1)=0.06; (1,3)=0.40	0.46
4	(4,1)=0.10; (2,2)=0.10	0.20
6	(3,2)=0.04; (2,3)=0.10	0.14
8	(4,2)=0.04	0.04

Check: $0.16+0.46+0.20+0.14+0.04 = 1.00$

Exercise 3.15

Hardware failures X (lab 1) and Y (lab 2). Joint distribution:

$X Y$	$Y=0$	$Y=1$	$Y=2$	$PX(x)$
$X=0$	0.52	0.20	0.04	0.76
$X=1$	0.14	0.02	0.01	0.17
$X=2$	0.06	0.01	0	0.07
$PY(y)$	0.72	0.23	0.05	1.00

(a) $P(\text{at least one failure})$

$$P(X+Y \geq 1) = 1 - P(X=0, Y=0) = 1 - 0.52 = 0.48$$

$P(\text{at least one failure}) = 0.48$

(b) Independence check

$$PX(0) * PY(0) = 0.76 * 0.72 = 0.5472 \neq P(0,0) = 0.52$$

X and Y are NOT independent.

Exercise 3.16

Hardware (X) and software (Y) failures: $P(0,0)=0.6$, $P(0,1)=0.1$, $P(1,0)=0.1$, $P(1,1)=0.2$.

$X Y$	$Y=0$	$Y=1$	$PX(x)$
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X=0	0.6	0.1	0.7
X=1	0.1	0.2	0.3
PY(y)	0.7	0.3	1.00

(a) Independence

$$P_X(0) * P_Y(0) = 0.7 * 0.7 = 0.49 \neq P(0, 0) = 0.60$$

NOT independent.

(b) E(X+Y) — expected total failures

$$E(X) = 0 * 0.7 + 1 * 0.3 = 0.3$$

$$E(Y) = 0 * 0.7 + 1 * 0.3 = 0.3$$

$$E(X+Y) = E(X) + E(Y) = 0.3 + 0.3 = 0.6$$

E(X+Y) = 0.6 failures per day

Exercise 3.17

Company A: \$10/share, gain/lose \$1 with prob 0.5. Company B: \$50/share, gain/lose \$1 with prob 0.5. X, Y = return per share. $E(X)=E(Y)=0$, $\text{Var}(X)=\text{Var}(Y)=1$.

Comparison of Portfolios (all cost \$1,000):

Portfolio	Composition	Cost	E(Profit)	Var(Profit)
(a)	100 shares A	$100 \times \$10 = \1000	0	$100^2 \times 1 = 10,000$
(b)	50 shares A + 10 shares B	$\$500 + \$500 = \$1000$	0	$50^2 \times 1 + 10^2 \times 1 = 2,600$
(c)	40 shares A + 12 shares B	$\$400 + \$600 = \$1000$	0	$40^2 \times 1 + 12^2 \times 1 = 1,744$

Portfolio (c) has the LOWEST risk (Var = 1,744 < 2,600 < 10,000).

Exercise 3.18

A: \$10/share, profit X%. B: \$50/share, profit Y%. Invest \$1,000.

$$E(X) = -3(0.3) + 0(0.2) + 3(0.5) = 0.6\%, \quad E(X^2) = 9(0.3) + 0 + 9(0.5) = 7.2, \quad \text{Var}(X) = 7.2 - 0.36 = 6.84$$

$$E(Y) = -3(0.4) + 3(0.6) = 0.6\%, \quad E(Y^2) = 9(0.4) + 9(0.6) = 9, \quad \text{Var}(Y) = 9 - 0.36 = 8.64$$

Portfolio	Dollar Profit	E(Profit)	Var(Profit)
(a) 100 shares A	$P = 10X$ (since \$1000 at X%)	$\$10 \times 0.6 = \6	$100 \times 6.84 = 684$
(b) 50A + 10B	$P = 5X + 5Y$	\$6	$25 \times 6.84 + 25 \times 8.64 = 387$
(c) 20 shares B	$P = 10Y$	\$6	$100 \times 8.64 = 864$

Least risky: (b) with Var = 387. Most risky: (c) with Var = 864.

Exercise 3.19

A: $P(X=2)=P(X=-2)=0.5 \Rightarrow E(X)=0, \text{Var}(X)=4$. B: $P(Y=4)=0.2, P(Y=-1)=0.8 \Rightarrow E(Y)=0, \text{Var}(Y)=4$.
(Verify: $E(Y)=4(0.2)+(-1)(0.8)=0.8-0.8=0$. $E(Y^2)=16(0.2)+1(0.8)=4$. $\text{Var}(Y)=4$.)

Strategy	Profit	E(Profit)	Var(Profit)
(a) 100 shares A	100X	0	$100^2 \times 4 = 40,000$
(b) 100 shares B	100Y	0	$100^2 \times 4 = 40,000$
(c) 50A + 50B	50X + 50Y	0	$50^2 \times 4 + 50^2 \times 4 = 20,000$

Portfolio (c) has the lowest risk (Var = 20,000 vs 40,000 for (a) and (b)).

Exercise 3.20

5% of computers defective, independently.

(a) P(exactly 3 defective in 20) — Binomial(20, 0.05)

$$\begin{aligned} P(X=3) &= C(20, 3) * 0.05^3 * 0.95^{17} \\ &= 1140 * 0.000125 * 0.4181 = 0.0596 \end{aligned}$$

P(X=3) $\approx$ 0.0596

(b) P(at least 5 tested to find 2 defective) — Negative Binomial(k=2, p=0.05)

$P(X \geq 5) = P(\text{fewer than 2 defective in first 4 trials}) = P(Y \leq 1)$ where $Y \sim \text{Binomial}(4, 0.05)$:

$$\begin{aligned} P(Y=0) &= 0.95^4 = 0.8145 \\ P(Y=1) &= C(4, 1) (0.05) (0.95)^3 = 4 * 0.05 * 0.8574 = 0.1715 \\ P(X \geq 5) &= P(Y \leq 1) = 0.8145 + 0.1715 = 0.9860 \end{aligned}$$

P(at least 5 tested) $\approx$ 0.9860

Exercise 3.21

20 computers, each infected with $p=0.4$. $X \sim \text{Binomial}(20, 0.4)$.

$$P(X \geq 10) = 1 - P(X \leq 9) = 1 - F(9)$$

Using Binomial table ($n=20, p=0.4$): $F(9) = 0.7553$.

$$P(X \geq 10) = 1 - 0.7553 = 0.2447$$

P(at least 10 computers infected) $\approx$ 0.2447

Exercise 3.22

16 parts, 5% defective. $X \sim \text{Binomial}(16, 0.05)$. $P(X > 3)$?

$$P(X=0) = 0.95^{16} = 0.4401$$

$$\begin{aligned}
P(X=1) &= C(16, 1) (0.05) (0.95)^{15} = 16 \cdot 0.05 \cdot 0.4633 = 0.3706 \\
P(X=2) &= C(16, 2) (0.05)^2 (0.95)^{14} = 120 \cdot 0.0025 \cdot 0.4877 = 0.1463 \\
P(X=3) &= C(16, 3) (0.05)^3 (0.95)^{13} = 560 \cdot 0.000125 \cdot 0.5133 = 0.0359 \\
F(3) &= P(X \leq 3) = 0.4401 + 0.3706 + 0.1463 + 0.0359 = 0.9929 \\
P(X > 3) &= 1 - 0.9929 = 0.0071
\end{aligned}$$

P(more than 3 defective) ≈ 0.0071

Exercise 3.23

Lecture canceled with $p=0.05$, independently.

(a) P(at least 4 of 15 classes canceled) — Binomial(15, 0.05)

$$P(X \geq 4) = 1 - F(3)$$

From Table A2 ($n=15$, $p=0.05$): $F(3) = 0.9945$

$$P(X \geq 4) = 1 - 0.9945 = 0.0055$$

P(at least 4 canceled) ≈ 0.0055

(b) P(10th class is the 3rd cancellation) — Negative Binomial($k=3$, $x=10$)

$$\begin{aligned}
P(X=10) &= C(9, 2) \cdot 0.05^3 \cdot 0.95^7 \\
&= 36 \cdot 0.000125 \cdot 0.6983 = 36 \cdot 0.0000873 = 0.00314
\end{aligned}$$

P(10th class is 3rd cancellation) ≈ 0.00314

Exercise 3.24

20% of internet sites contain the keyword.

(a) P(at least 5 of first 10 sites contain keyword) — Binomial(10, 0.2)

$$P(X \geq 5) = 1 - F(4)$$

From Table A2 ($n=10$, $p=0.2$): $F(4) = 0.9672$.

$$P(X \geq 5) = 1 - 0.9672 = 0.0328$$

P(at least 5 contain keyword) ≈ 0.0328

(b) P(at least 5 sites visited to find first keyword) — Geometric($p=0.2$)

$$P(X \geq 5) = (1-0.2)^4 = 0.8^4 = 0.4096$$

P(at least 5 sites visited) ≈ 0.4096

Exercise 3.28

Messages arrive at average rate 9/hour. $X \sim \text{Poisson}(\lambda=9)$.

(a) P(at least 5 messages in next hour)

$$P(X \geq 5) = 1 - F(4)$$

From Table A3 ($\lambda=9$): $F(4) = 0.0550$.

$$P(X \geq 5) = 1 - 0.0550 = 0.9450$$

P(at least 5 messages) ≈ 0.9450

(b) P(exactly 5 messages in next hour)

$$P(X=5) = F(5) - F(4) = 0.1157 - 0.0550 = 0.0607$$

Direct formula check: $P(X=5) = e^{(-9)} * 9^5 / 5! = 0.001234 * 59049 / 120 \approx 0.0607$

P(exactly 5 messages) ≈ 0.0607

CHAPTER 4 — Continuous Distributions

Exercise 4.1

Lifetime density: $f(x) = k/x^4$ for $x \geq 1$, else 0.

Find k

$$\begin{aligned} \text{Integral from 1 to inf: } k/x^4 \, dx &= k * [-1/(3x^3)] \text{ from 1 to inf} = k/3 \\ &= 1 \\ \Rightarrow k &= 3 \end{aligned}$$

CDF F(x)

$$\begin{aligned} F(x) &= \text{Integral from 1 to x of } 3/t^4 \, dt = [-1/t^3] \text{ from 1 to x} = 1 - 1/x^3, \quad x \geq 1 \\ F(x) &= 0 \quad \text{for } x < 1 \end{aligned}$$

P(lifetime > 2 years)

$$P(X > 2) = 1 - F(2) = 1 - (1 - 1/8) = 1/8 = 0.125$$

k = 3; F(x) = 1 - 1/x^3 (x ≥ 1); P(X > 2) = 0.125

Exercise 4.2

Reboot time density: $f(x) = C(10-x)^2$ for $0 < x < 10$.

(a) Find C

$$\begin{aligned} \text{Integral from 0 to 10: } C(10-x)^2 \, dx &= C * [-(10-x)^3/3] \text{ from 0 to 10} \\ &= C * (0 - (-1000/3)) = 1000C/3 = 1 \\ \Rightarrow C &= 3/1000 = 0.003 \end{aligned}$$

C = 3/1000 = 0.003

(b) P(1 < X < 2)

$$\begin{aligned} P(1 < X < 2) &= \text{Integral from 1 to 2 of } C(10-x)^2 \, dx \\ &= (3/1000) * [-(10-x)^3/3] \text{ from 1 to 2} \\ &= (3/1000) * (-8^3/3 + 9^3/3) \\ &= (3/1000) * (-512/3 + 729/3) \\ &= (3/1000) * (217/3) = 217/1000 = 0.217 \end{aligned}$$

P(1 < X < 2) = 0.217

Exercise 4.3

Installation time density: $f(x) = k(1 - x^3)$ for $0 < x < 1$.

Find k

Integral from 0 to 1: $k(1-x^3) dx = k[x - x^4/4]$ from 0 to 1 = $k(1 - 1/4) = 3k/4 = 1$
 $\Rightarrow k = 4/3$

$P(X < 1/2)$

$P(X < 1/2) = \text{Integral from 0 to } 1/2 \text{ of } (4/3)(1-x^3) dx$
 $= (4/3) * [x - x^4/4] \text{ from 0 to } 1/2$
 $= (4/3) * [1/2 - (1/2)^4/4]$
 $= (4/3) * [1/2 - 1/64]$
 $= (4/3) * (32-1)/64 = (4/3) * (31/64) = 31/48 = 0.6458$

$k = 4/3$; $P(X < 1/2) = 31/48 \approx 0.6458$

Exercise 4.4

Hardware lifetime density: $f(x) = K - x/50$ for $0 < x < 10$, else 0.

(a) Find K

Integral from 0 to 10: $(K - x/50) dx = [Kx - x^2/100]$ from 0 to 10 = $10K - 1 = 1$
 $\Rightarrow 10K = 2 \Rightarrow K = 0.2$

$K = 0.2$

(b) $P(\text{failure within 5 years})$

$P(X < 5) = \text{Integral from 0 to 5 of } (0.2 - x/50) dx$
 $= [0.2x - x^2/100] \text{ from 0 to 5}$
 $= 0.2(5) - 25/100 = 1.0 - 0.25 = 0.75$

$P(X < 5) = 0.75$

(c) Expected lifetime $E(X)$

$E(X) = \text{Integral from 0 to 10 of } x(0.2 - x/50) dx$
 $= \text{Integral from 0 to 10 of } (x/5 - x^2/50) dx$
 $= [x^2/10 - x^3/150] \text{ from 0 to 10}$
 $= 100/10 - 1000/150 = 10 - 20/3 = 10/3 = 3.333 \text{ years}$

$E(X) = 10/3 \approx 3.333 \text{ years}$

Exercise 4.5

Joint density $f(x,y) = C(x^2 + y)$ for $-1 \leq x \leq 1$, $0 \leq y \leq 1$.

(a) Find C

Double integral: $C * \text{Integral from } -1 \text{ to } 1 \text{ Integral from } 0 \text{ to } 1 \text{ of } (x^2 + y) dy dx$
Inner integral: $[x^2*y + y^2/2] \text{ from } 0 \text{ to } 1 = x^2 + 1/2$
Outer integral: $[x^3/3 + x/2] \text{ from } -1 \text{ to } 1 = (1/3 + 1/2) - (-1/3 - 1/2)$
 $= 2(5/6) = 5/3$
So $C * (5/3) = 1 \Rightarrow C = 3/5$

$$C = 3/5 = 0.6$$

(b) Marginal densities & Independence

$$f_X(x) = \text{Integral from } 0 \text{ to } 1 \text{ of } (3/5)(x^2+y) \, dy = (3/5)(x^2 + 1/2), \quad -1 \leq x \leq 1$$

$$\begin{aligned} f_Y(y) &= \text{Integral from } -1 \text{ to } 1 \text{ of } (3/5)(x^2+y) \, dx = (3/5)[x^3/3+yx] \text{ from } -1 \text{ to } 1 \\ &= (3/5)[(1/3+y)-(-1/3-y)] = (3/5)(2/3+2y) = (2/5)(1+3y), \quad 0 \leq y \leq 1 \end{aligned}$$

Independence? $f(x,y)$ must equal $f_X(x)*f_Y(y)$ for all x,y .

$$f_X(x)*f_Y(y) = (3/5)(x^2+1/2) * (2/5)(1+3y) \neq (3/5)(x^2+y) \text{ in general.}$$

X and Y are NOT independent.

(c) Probabilities

$$\begin{aligned} P(Y < 0.6) &= \text{Integral from } 0 \text{ to } 0.6 \text{ of } (2/5)(1+3y) \, dy \\ &= (2/5)[y + 3y^2/2] \text{ from } 0 \text{ to } 0.6 \\ &= (2/5)[0.6 + 3(0.36)/2] = (2/5)(0.6 + 0.54) = (2/5)(1.14) = 0.456 \end{aligned}$$

For $P(Y < 0.6 \mid X < 0.5)$, compute joint and marginal probabilities:

$$P(Y < 0.6, X < 0.5) = \text{Integral from } -1 \text{ to } 0.5 \text{ Integral from } 0 \text{ to } 0.6 \text{ of } (3/5)(x^2+y) \, dy \, dx$$

$$\text{Inner integral: } (3/5)[0.6x^2 + 0.18]$$

$$\text{Outer integral: } (3/5)[0.2x^3 + 0.18x] \text{ from } -1 \text{ to } 0.5$$

$$= (3/5)[(0.025+0.09)-(-0.2-0.18)] = (3/5)[0.115+0.38] = (3/5)(0.495) = 0.297$$

$$P(X < 0.5) = \text{Integral from } -1 \text{ to } 0.5 \text{ of } (3/5)(x^2+1/2) \, dx$$

$$= (3/5)[x^3/3+x/2] \text{ from } -1 \text{ to } 0.5$$

$$= (3/5)[(0.0417+0.25)-(-0.333-0.5)] = (3/5)[0.2917+0.833] = (3/5)(1.125) = 0.675$$

$$P(Y < 0.6 \mid X < 0.5) = 0.297/0.675 = 0.44$$

$$P(Y < 0.6) = 0.456; P(Y < 0.6 \mid X < 0.5) = 0.297/0.675 \approx 0.44$$

Exercise 4.16

Standard Normal Z. (Using Table A4: $\Phi(1.25) = 0.8944$)

Part	Expression	Calculation	Answer
(a)	$P(Z < 1.25)$	$\Phi(1.25)$	0.8944
(b)	$P(Z \leq 1.25)$	Same as (a) for continuous	0.8944
(c)	$P(Z > 1.25)$	$1 - \Phi(1.25)$	0.1056
(d)	$P(Z \leq 1.25)$	$2*\Phi(1.25) - 1$	0.7888
(e)	$P(Z < 6.0)$	$\Phi(6.0)$	≈ 1.0000
(f)	$P(Z > 6.0)$	$1 - \Phi(6.0)$	≈ 0.0000

(g)	$P(Z \leq z) = 0.80$	$z = \Phi^{-1}(0.80)$	$z \approx 0.842$
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Exercise 4.17

Standard Normal Z. (Using Table A4: $\Phi(0.99) = 0.8389$)

Part	Expression	Calculation	Answer
(a)	$P(Z \geq 0.99)$	$1 - \Phi(0.99)$	0.1611
(b)	$P(Z \leq -0.99)$	$\Phi(-0.99) = 1 - \Phi(0.99)$	0.1611
(c)	$P(Z < 0.99)$	$\Phi(0.99)$	0.8389
(d)	$P(Z > 0.99)$	$2(1 - \Phi(0.99))$	0.3222
(e)	$P(Z < 10.0)$	$\Phi(10.0)$	≈ 1.0000
(f)	$P(Z > 10.0)$	$1 - \Phi(10.0)$	≈ 0.0000
(g)	$P(Z < z) = 0.90$	$z = \Phi^{-1}(0.90)$	$z \approx 1.282$

Exercise 4.18

Normal X: $E(X) = -3$, $\text{Var}(X) = 4$, $\sigma = 2$. Standardize: $Z = (X+3)/2$.

Part	Standardization	Calculation	Answer
(a) $P(X \leq 2.39)$	$Z = (2.39+3)/2 = 2.695$	$\Phi(2.70)$	≈ 0.9965
(b) $P(Z \geq -2.39)^*$	Direct: $P(Z \geq -2.39)$	$1 - \Phi(-2.39) = \Phi(2.39)$	0.9916
(c) $P(X \geq 2.39)$	$P(X \leq -2.39) + P(X \geq 2.39)$	$\Phi(0.305) + [1 - \Phi(2.695)]$	$0.6198 + 0.0035 = 0.6233$
(d) $P(X+3 \geq 2.39)$	$ X+3 = 2 Z $, so $P(Z \geq 1.195)$	$2(1 - \Phi(1.195))$	$2(0.116) = 0.232$
(e) $P(X < 5)$	$Z = (5+3)/2 = 4.0$	$\Phi(4)$	≈ 1.0000
(f) $P(X < 5)$	$P(-5 < X < 5)$	$\Phi(4) - \Phi(-1)$	0.8413
(g) $P(X > x) = 0.33$	$(x+3)/2 = \Phi^{-1}(0.67) = 0.44$	$x = 2(0.44) - 3$	$x = -2.12$

* (b) uses the standardized Z notation directly from the problem.

Exercise 4.19

$P(|X - \mu| \leq 3\sigma)$ — proportion conforming within 3 standard deviations.

(a) Normal(μ , σ)

$$P(|X - \mu| \leq 3\sigma) = P(-3 \leq Z \leq 3) = \Phi(3) - \Phi(-3) = 0.9974 - 0.0026 = 0.9974$$

Normal: 99.74% of items conform.

(b) Uniform(a, b)

For Uniform: $\sigma = (b-a)/\sqrt{12}$. The 3-sigma interval has half-width:

$$3\sigma = 3(b-a)/\sqrt{12} = (b-a)\sqrt{3}/2 \approx 0.866(b-a)$$

The full range of Uniform is $(b-a)$. Since $0.866(b-a) < (b-a)/2$, wait: 3σ half-width $= 0.866(b-a) > (b-a)/2 = 0.5(b-a)$. So the 3-sigma interval extends BEYOND $[a,b]$. Thus:

Uniform: 100% of items fall within 3 standard deviations.

Exercise 4.20

$P(|X - \mu| > 1.5\sigma)$ — proportion falling beyond 1.5 standard deviations.

(a) Normal(μ , σ)

$$P(|Z| > 1.5) = 2(1 - \Phi(1.5)) = 2(1 - 0.9332) = 2(0.0668) = 0.1336$$

Normal: 13.36% fall beyond 1.5 standard deviations.

(b) Uniform(a , b)

$$1.5\sigma = 1.5(b-a)/\sqrt{12} = (b-a)1.5/(2\sqrt{3}) = (b-a)0.4330.$$

The Uniform half-range is $(b-a)/2 = 0.5(b-a)$. Since $0.4330 < 0.5$, the interval $[\mu - 1.5\sigma, \mu + 1.5\sigma]$ lies strictly within $[a,b]$.

$$P(X < \mu - 1.5\sigma) = (\mu - 1.5\sigma - a)/(b-a) = (0.5 - 0.4330) = 0.0670$$

$$P(|X - \mu| > 1.5\sigma) = 2 * 0.0670 = 0.1340 = 13.40\%$$

Uniform: 13.40% fall beyond 1.5 standard deviations.

Exercise 4.21

NBA players: $\mu = 79$ inches (6'7"), $\sigma = 3.89$ inches. Normal distribution.

(a) $P(\text{height} > 7 \text{ feet} = 84 \text{ inches})$

$$Z = (84 - 79)/3.89 = 5/3.89 = 1.285$$

$$P(X > 84) = 1 - \Phi(1.285) = 1 - \Phi(1.29) \approx 1 - 0.9015 = 0.0985$$

$\approx 9.85\%$ of players are taller than 7 feet.

(b) Height threshold for tallest 20%

$$P(Z > z_0) = 0.20 \Rightarrow \Phi(z_0) = 0.80 \Rightarrow z_0 = \Phi^{-1}(0.80) = 0.842$$

$$h = \mu + z_0\sigma = 79 + 0.842*3.89 = 79 + 3.275 = 82.28 \text{ inches} = 6'10.3"$$

Top 20% of players are at least 82.3 inches (6'10.3") tall.

Exercise 4.22

Normal household income: $\mu = 900$, $\sigma = 200$ coins.

(a) $P(\text{income} < 640)$ — qualifies for free milk

$$Z = (640 - 900)/200 = -260/200 = -1.30$$

$$P(X < 640) = \Phi(-1.30) = 0.0968$$

About 9.68% of households qualify for free milk.

(b) Income threshold for lowest 5% — free sandwiches

$$P(X \leq x) = 0.05 \Rightarrow (x - 900)/200 = \Phi^{-1}(0.05) = -1.645$$

$$x = 900 + 200*(-1.645) = 900 - 329 = 571 \text{ coins}$$

Households with income below 571 coins receive free sandwiches.

Exercise 4.23

Component lifetime: $E(X) = 5000$ hrs, $\sigma = 100$ hrs. Sample $n = 400$.

By CLT, the sample average $\bar{X} \sim \text{Normal}(\mu=5000, \sigma_{\bar{X}} = 100/\sqrt{400} = 5)$.

$$P(\bar{X} < 5012) = P(Z < (5012 - 5000)/5) = P(Z < 2.40)$$

$$= \Phi(2.40) = 0.9918$$

$P(\text{average lifetime} < 5012 \text{ hrs}) \approx 0.9918$

Exercise 4.24

82 files, $E(T_i) = 15$ sec, $\text{Var}(T_i) = 16 \text{ sec}^2$. Total time $S = \text{sum of } T_i$.

$$E(S) = 82 * 15 = 1230 \text{ sec}$$

$$\text{Var}(S) = 82 * 16 = 1312 \text{ sec}^2 \quad \text{Std}(S) = \sqrt{1312} = 36.22 \text{ sec}$$

20 minutes = 1200 seconds.

$$P(S < 1200) = P(Z < (1200 - 1230)/36.22) = P(Z < -0.83)$$

$$= \Phi(-0.83) = 0.2033$$

$P(\text{software installed in} < 20 \text{ min}) \approx 0.2033$

Exercise 4.25

400 chips, $p = 0.06$ defective. $X \sim \text{Binomial}(400, 0.06)$. $\mu = 24$, $\sigma = \sqrt{400*0.06*0.94} = \sqrt{22.56} = 4.75$.

(a) $P(20 \leq X \leq 25)$ [with continuity correction: $P(19.5 < X < 25.5)$]

$$P(19.5 < X < 25.5) = P((19.5-24)/4.75 < Z < (25.5-24)/4.75)$$

$$= P(-0.947 < Z < 0.316)$$

$$= \Phi(0.316) - \Phi(-0.947)$$

$$= 0.6241 - 0.1719 = 0.4522$$

$P(20 \leq X \leq 25) \approx 0.4522$

(b) $P(\text{at least 8 of 40 inspectors find 20-25 defective chips})$

Each inspector independently finds 20-25 defectives with $p = 0.4522$. $W \sim \text{Binomial}(40, 0.4522)$. $\mu_W = 18.09$, $\sigma_W = \sqrt{40*0.4522*0.5478} = 3.148$.

$$P(W \geq 8) = P(Z \geq (7.5 - 18.09)/3.148) \quad [\text{continuity correction}]$$

$$= P(Z \geq -3.364) = \Phi(3.364) \approx 0.9996$$

$P(\text{at least 8 inspectors find 20-25 defective}) \approx 0.9996$

Exercise 4.26

80 images: $E(\text{size}) = 0.6$ MB, $\sigma = 0.4$ MB. Total $S = \text{sum of sizes}$.

```
E(S) = 80 * 0.6 = 48 MB
Var(S) = 80 * (0.4)^2 = 80 * 0.16 = 12.8 MB^2    Std(S) = sqrt(12.8)
= 3.578 MB
P(47 < S < 50) = P((47-48)/3.578 < Z < (50-48)/3.578)
= P(-0.279 < Z < 0.559)
= Phi(0.559) - Phi(-0.279)
= Phi(0.56) - Phi(-0.28) = 0.7123 - 0.3897 = 0.3226
```

P(47 MB < total size < 50 MB) ≈ 0.3226

End of Solutions — Chapters 3 & 4